

Looping Structures 2

for loops

Maze

In order to visit every room in this hallway

1	■	()	■	■	■	■
2	■		■	■	■	■
3	■		■	■	■	■

And given a list of rooms ["A", "B", "C", "D"] ,

What would the program look like, if we already know exactly how many rooms there are?

For loops

A `for` loop is a style of loop which goes through each item in a sequence, one at a time

```
1  weather_forecast = ["sunny", "rainy", "cloudy", "sunny"]
2
3  for day in weather_forecast:
4      print("Today is", day)
5
6      if day == "rainy":
7          print("Bring an umbrella!")
```

Lists

A `list` is just a sequence of items, in order, that we can loop through

```
1  fruits = ["apple", "banana", "cherry"]
2  scores = [85, 92, 78, 100]
3  mixed = ["Alice", 25, True]
```

For loops go through a list one item at a time, in order, from start to end

```
1  for fruit in fruits:
2      print(fruit)
```

Ranges

A `range` is a quick way to generate a sequence of numbers, without typing them all out

```
1 range(5)           # 0, 1, 2, 3, 4
2 range(1, 6)        # 1, 2, 3, 4, 5
3 range(10, 0, -1)    # 10, 9, 8, ... 1
4 range(0, 20, 2)     # 0, 2, 4, ... 18
```

We usually use `range` with a `for` loop, instead of typing out a whole list of numbers

```
1 for number in range(1, 6):
2     print(number)
```

For loop examples

Counting up from 1 to 5

```
1  for count in range(1, 6):  
2      print(count)
```

For loop examples

Counting down from 10 to 1

```
1  for count in range(10, 0, -1):  
2      print(count)
```

For loop examples

Summing all numbers in a list

```
1 numbers = [4, 8, 15, 16, 23, 42]
2 total = 0
3
4 for number in numbers:
5     total = total + number
6
7 print("Total:", total)
```


For loop examples

Printing every other letter in a word

```
1 word = "PROGRAMMING"  
2  
3 for letter in word[::2]:  
4     print(letter)
```

For loop examples

Doubling each value in a list

```
1 values = [1, 2, 4, 8]
2
3 for value in values:
4     print(value * 2)
```

Break and Continue Statements

In a for loop it's often useful to *skip* an item, without ending

For example,

```
1  for number in range(1, 11):  
2      if number % 2 == 0:  
3          continue  
4      print(number)
```

`continue` skips straight to the next item in the sequence, while `break` stops the loop completely

For loop with break examples

Searching a list for a target value

```
1  numbers = [4, 8, 15, 16, 23, 42]
2  target = 16
3  found = False
4
5  for number in numbers:
6      if number == target:
7          found = True
8          break
9
10 print("Found:", found)
```

For loop with break examples

Stopping as soon as a total goes over a limit

```
1  numbers = [10, 20, 30, 40, 50]
2  total = 0
3
4  for number in numbers:
5      total = total + number
6
7      if total > 60:
8          break
9
10 print("Total stopped at:", total)
```

For loop with continue examples

Skipping negative numbers when summing a list

```
1  numbers = [5, -3, 8, -1, 4]
2  total = 0
3
4  for number in numbers:
5      if number < 0:
6          continue
7      total = total + number
8
9  print("Total:", total)
```

For loop with continue examples

Printing only the vowels in a word

```
1 word = "ELEPHANT"
2 vowels = "AEIOU"
3
4 for letter in word:
5     if letter not in vowels:
6         continue
7     print(letter)
```

While loop vs For loop

Use a `for` loop when you know how many times to loop, or you are going through a sequence

Use a `while` loop when you don't know how many times to loop, and are waiting for a condition to change

```
1  # for loop: known number of items
2  for student in class_list:
3      print(student)
4
5  # while loop: unknown number of tries
6  while guess != secret_number:
7      guess = int(input("Guess the number: "))
```


Exercises

Using a `for` loop, write a program that:

1. Prints every even number from 2 to 20
2. Goes through a list of names and prints a greeting for each one
3. Starts at 100 and counts down by 7 each time until 0 or below, printing each step
4. Adds up numbers from 1 to 50 and prints the final total
5. Goes through a list of numbers entered by the user, stopping as soon as it finds a negative number, and prints how many numbers were checked before stopping

Exercises

1. A bank account starts with \$50 and earns \$5 interest every year. Print the balance for each of the first 10 years.
2. A car has 12 litres of fuel and uses 3 litres per hour of driving. Print the remaining fuel after each hour for a 6 hour trip.
3. A student volunteers 4 hours a week for 15 weeks. Print how many total hours they have after each week.
4. You are given a list of numbers. Print only the numbers that are divisible by 3, skipping all others.
5. A plant is 2cm tall and grows 3cm every day. Print the height each day for 14 days, and stop early if it grows taller than 30cm.